

Model Predictive Control Strategy Based on Differential Discrete Particle Swarm Optimization

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Abstract— A model-based predictive control strategy, aided by a differential discrete particle swarm optimization, is proposed. In particular, the proposed approach extends the traditional discrete binary version of the particle swarm algorithm by re-designing the particle section (all possible solutions in the search space) in order to represent a sequence of discrete controls. In this way, the "velocity section" turns out to be related to the probability of achieving the suitable discrete control value. As an experimental case study, this strategy is applied to a temperature control in a building aimed at energy saving. The proposed method is compared with a standard particle swarm algorithm, and experimental results are discussed.

Keywords: Artificial Intelligence, Predictive Control, Energy Optimal Control.

I. INTRODUCTION

In a civil building, the climate control system should maximize the trade-off between the comfort level for people living and working and the energy saving. Conventional control techniques achieve this goal by regulating the air room temperature only through the feedback from the user or from a pre-defined reference indoor temperature. In most cases, the external environment (i.e. outdoor temperature, humidity and solar radiation) can nullify rapidly the control action, by causing a significant waste of energy.

Model predictive control (MPC) [1] has the advantage of predicting the behaviour of dependent variables (output) of the controlled system or process, with respect to changes in the process independent variables (inputs). Thus, the system is able to react before any deviations in the controlled variable take place, by avoiding delays in the system response [2]. If applied to climate control in a building, this approach leads to an effective management of all the variables in the system (indoor temperature, outdoor temperature, humidity and solar

radiation) and, in turn, to saving energy, such as shown experimentally in [3-6].

For these reasons, a Model Predictive Controller (MPC) for temperature regulation of a building, where people can change the desired temperature of the room in order to provide the best environmental conditions turns out to be useful. MPC is an on-line constrained optimization method [1], based on an iterative receding horizon control strategy. The aim of the control is to minimize energy consumption, while keeping the state variables as close as possible to the desired one.

However, the prerogative of this class of algorithms is to employ models able to predict several variables over a specified horizon. On this basis, an optimization problem is defined in order to find the best controls satisfying the model and the cost function. Model predictive control cost functions often define a very complex, non linear, non convex search space, therefore turn out to be suitable for heuristic approaches, such as in [3], where in a real-time application, a Particle Swarm Optimization (PSO) [7] searches for optimal control inputs over a specified prediction horizon.

In this paper, for a temperature control task with a discrete search space for possible control assignments, a novel Differential Discrete Particle Swarm Optimization (DDPSO) for MPC is proposed. In particular, the proposed approach extends the work in [8], by re-designing the *particle section* (all possible solutions in the search space) in order to represent a sequence of discrete controls. In this way, the "velocity section" turns out to be related to the probability of achieving the suitable discrete control value. In this paper, the proposed approach is validated and compared with a standard PSO on an experimental case study and the results are discussed.

II. THE PROPOSED METHOD

In this Section, first, an overview of a standard Particle Swarm Optimization is given, then, the proposed Differential Discrete Particle Swarm method is presented and, finally the basic MPC control is explained.

A. Particle Swarm Optimization

PSO is a population-based stochastic optimization technique [7] inspired by the social behaviour of bird flocking. The system is initialized with random solutions ("particles") with an assigned random "velocity" v_i , and then "flown" through the parameter space u_i . Each i -th particle (i.e. a potential solution) "flies" within its search space, conditioned by values of its *personal best position* pb (i.e., the best position found by the particle so far, and having the highest scoring according to a fitness function), and the *global best position* G (i.e. the best position found by all other particles). As a result, each particle searches around a region defined by its personal and all its neighbours best positions.

In the PSO algorithm, v_i and u_i , for $i = 1, \dots, n$, at each step k , are updated according to the following two equations:

$$\begin{aligned} v_i &= v_i(k) + c_1^1 (pb_i - u_i) + c_1^2 (G - u_i) \\ u_i &= u_i + v_i \end{aligned} \quad (1)$$

The first equation update the "velocities" of the particles while the second is relative to the "positions" of them. The equations highlight that the possible values taken by the position variables vary continuously in R .

On the contrary, in the proposed approach, such as detailed in next Section, the second equations is suitably "discretized" in order to better fit the context of discrete-space of control solutions.

B. The Differential Discrete Particle Swarm method

In temperature control task, the position space is the discrete space of possible control assignments, thus in the proposed approach, the particle section has to be redesigned suitably to represent a sequence of discrete controls. The "velocity section" (Fig 1) is related to the probability of having a suitable discrete control value. By extending the binary discrete PSO in [8], an unchanged equation for the "velocity section" update is proposed, though differences occur in the "position section" (Fig. 1). Given a population P and a set of discrete controls $U = \{u_1, \dots, u_s\}$, the steps of the proposed DDPSO are shown in Fig. 1.

The process is tuned by the acceleration constants $c^1 = w_1 R_1$ and $c^2 = w_2 R_2$, where R_1 and R_2 are random variables in $[0,1]$, and w_1 and w_2 represent the weighting of the stochastic acceleration terms pulling each particle towards *Global* and *personal* best solutions.

The step involving the random rnd value represents the stochastic choice of the control driven by the "velocity section".

At step k

for each element i of P

pick a random $u^{proposed} \in U$

(a) compute "velocity section"

$$v_i^{proposed} = v_i(k) + c_i^1 (pb_i - u_i^{proposed}) + c_i^2 (G - u_i^{proposed})$$

$$v_i^{previous} = v_i(k) + c_i^1 (pb_i - u_i(k)) + c_i^2 (G - u_i(k))$$

(b) compute "position section" vector

$$ratio = v_i^{proposed} / v_i^{previous}$$

$$u_i(k+1) = \begin{cases} u_i^{previous} & \text{if } rnd < \sigma(ratio) \\ u_i^{proposed} & \text{otherwise} \end{cases}$$

Figure 1: the proposed DDPSO procedure. It mainly consists in two section: (a) "velocity section" and (b) "position section"

The overall process optimization is guided by a cost function, minimizing the energy consumption of the control system.

A general form of the cost function is generally given by two terms, one controlling the distance from the reference weighted by a parameter λ_1 , and the other one trying to minimize the control effort in order to achieve the desired target, weighted by a second parameter λ_2 :

$$E(\mathbf{u}) = \lambda_1 \sum_{i=1}^s (\mathbf{tg}(t_i) - \mathbf{x}(t_i))^2 + \lambda_2 \sum_{i=1}^s \Delta u(i) \quad (2)$$

The tuning between these two terms drives the solutions to balance the accuracy of the desired target and, in this case, the energy consumption of the control architecture. The proposed cost function is redefined in order to achieve the controls, suitably satisfying the Room Temperature Control, able to perform the Energy consumption.

$$E(\mathbf{u}) = \lambda_1 \sum_{i=1}^s (\mathbf{tg}(t_i) - \mathbf{x}(t_i))^2 + \lambda_2 \sum_{i=1}^s |u(t_i)|^2 \quad (3)$$

C. Model Predictive Control

The performance of MPC relies on two main modules: a *Forward* model, predicting the next state variables $\mathbf{x}$ of the process on the basis of the receding horizon, and an *Inverse* model, trying to optimize a sequence of future control values $\mathbf{u}$ **where** the system output is expected to track a given reference input $\mathbf{tg}$ (Fig. 2).

The *Forward* model is identified previously from process data in order to minimize prediction mismatches. The estimated model has to perform in real time, thus has to be as simple as possible and capable of describing the system dynamics in order to predict future outputs accurately.

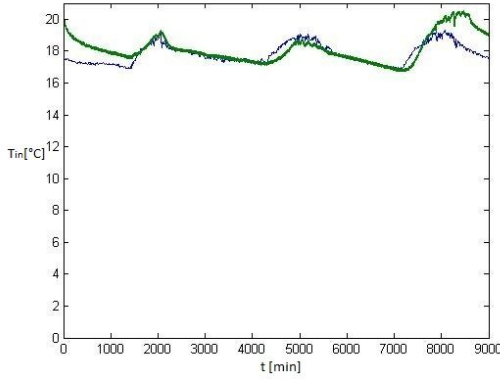


Figure 3: Comparison between measured outdoor temperature (green line) and simulated data (blue line) by means of model (7).

Parameter setting		
Parameter	PSO	DDPSO
W_1	2	2
W_2	2	2
Number of particles	24	24

Table 1: Settings for the comparison tests between PSO and the proposed approach.

The proposed approach achieves better solutions in a number of iterations less than the standard PSO. This result is particularly useful for MPC control in real-time applications, where a good solution has to be achieved in a bounded time. In Tab. 2, the obtained results are shown, while in Fig. 4, the mean values of E in (3) over the 500 iterations for both the algorithms are reported.

Performance Comparison		
Performance index	DDPSO	PSO
E (mean value)	12.93	336.90
SP_e (mean value)	0.73	26.92
E_v (mean value)	12.49	310.75
σ_E	0.35	19.37
σ_{EV}	0.41	15.68
σ_{SP_e}	0.36	11.67

Table 2: Experimental mean and standard deviation on 20 runs results for the proposed DDPSO and the standard PSO.

IV. CONCLUSIONS

A model predictive control approach for air temperature controller in a civil building has been presented. After a preliminary tuning for identifying the forward model of the physical system, a differential discrete particle swarm

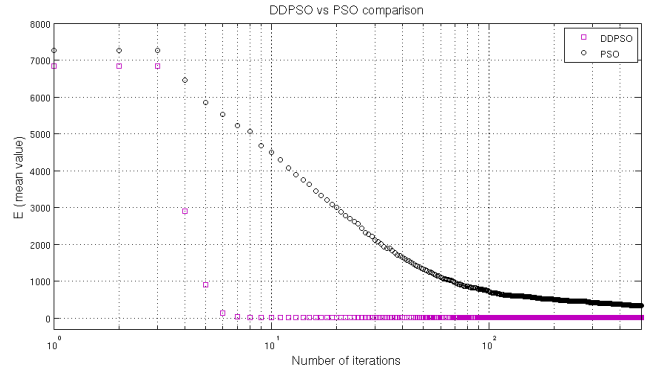


Figure 4: Comparison between the proposed DDPSO and a standard PSO.

optimization method (DDPSO) is developed in order to solve the specific discrete-controls prediction problem to be faced. The proposed DDPSO method for MPC control is then compared with a standard particle swarm optimization algorithm. Preliminary experimental results, with encouraging performance of the proposed DDPSO compared with the standard PSO, suggest future applications in context where a fast real-time reaction is needed.

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